

# PDF annotation

## Intro

Tables and plots to answer our research questions:

- RQ1a: Which theoretical concepts are being measured in opinion mining?
- RQ1b: Which methods and measurements are used for opinion mining?
- RQ1c: Which tools are used for opinion mining?
- RQ1d: Which data is used in CSS opinion mining?
- RQ2: Are there disciplinary differences in concepts, methods, tools, and data in opinion mining?
- RQ3: Do concepts, methods, tools, and data in opinion mining change over time?

## Packages and functions

## Data

## Main

## Validation

## RQ1a theoretical concepts

Question Q4\_1\_Opinion-Evaluation

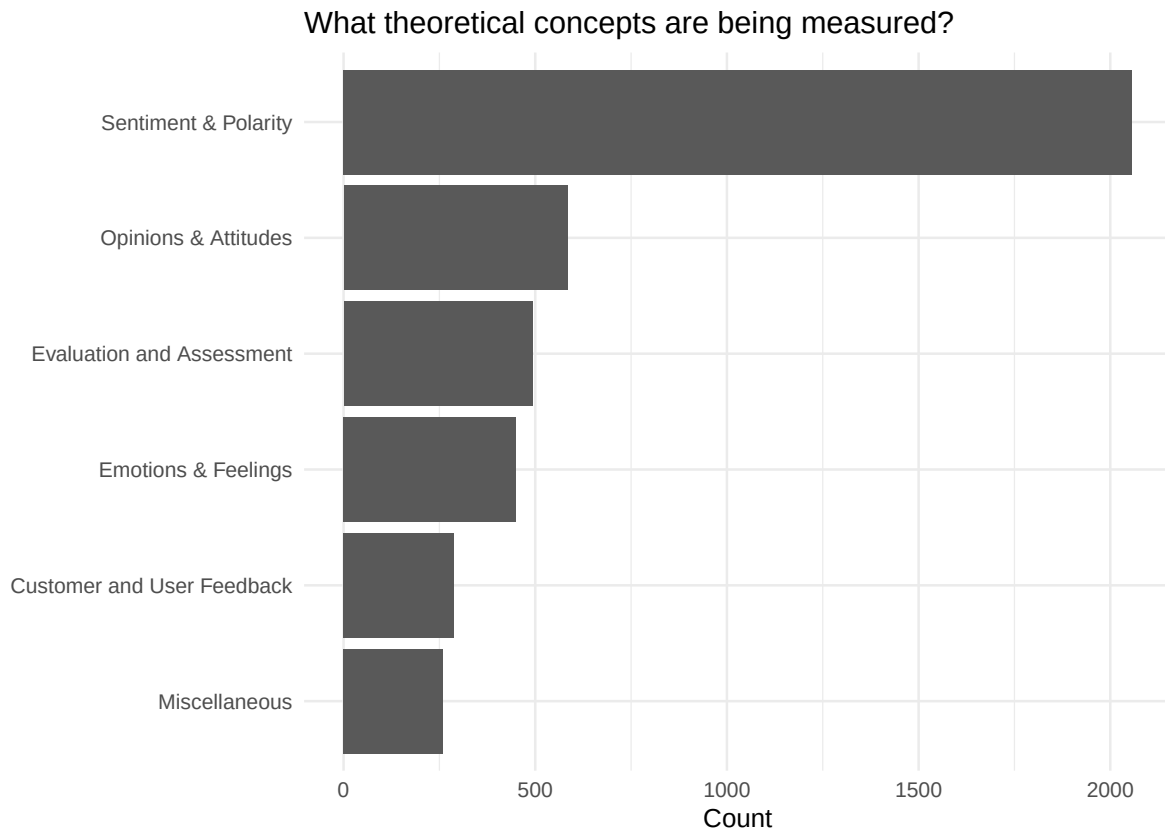


Figure 1: Categories of Opinion Evaluation Concepts

## RQ1b methods and measurements used for opinion mining?

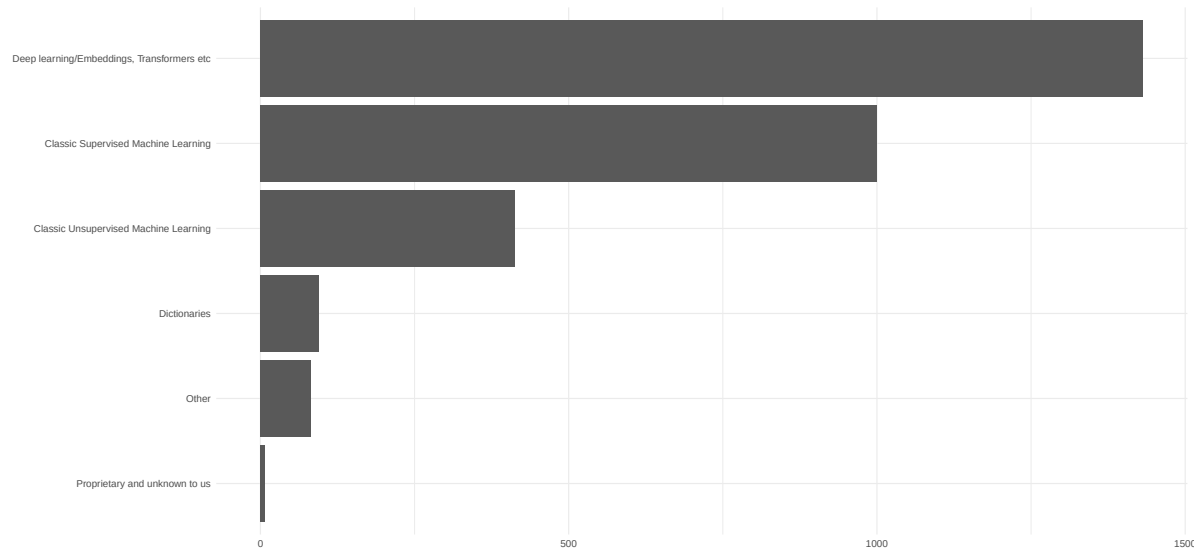
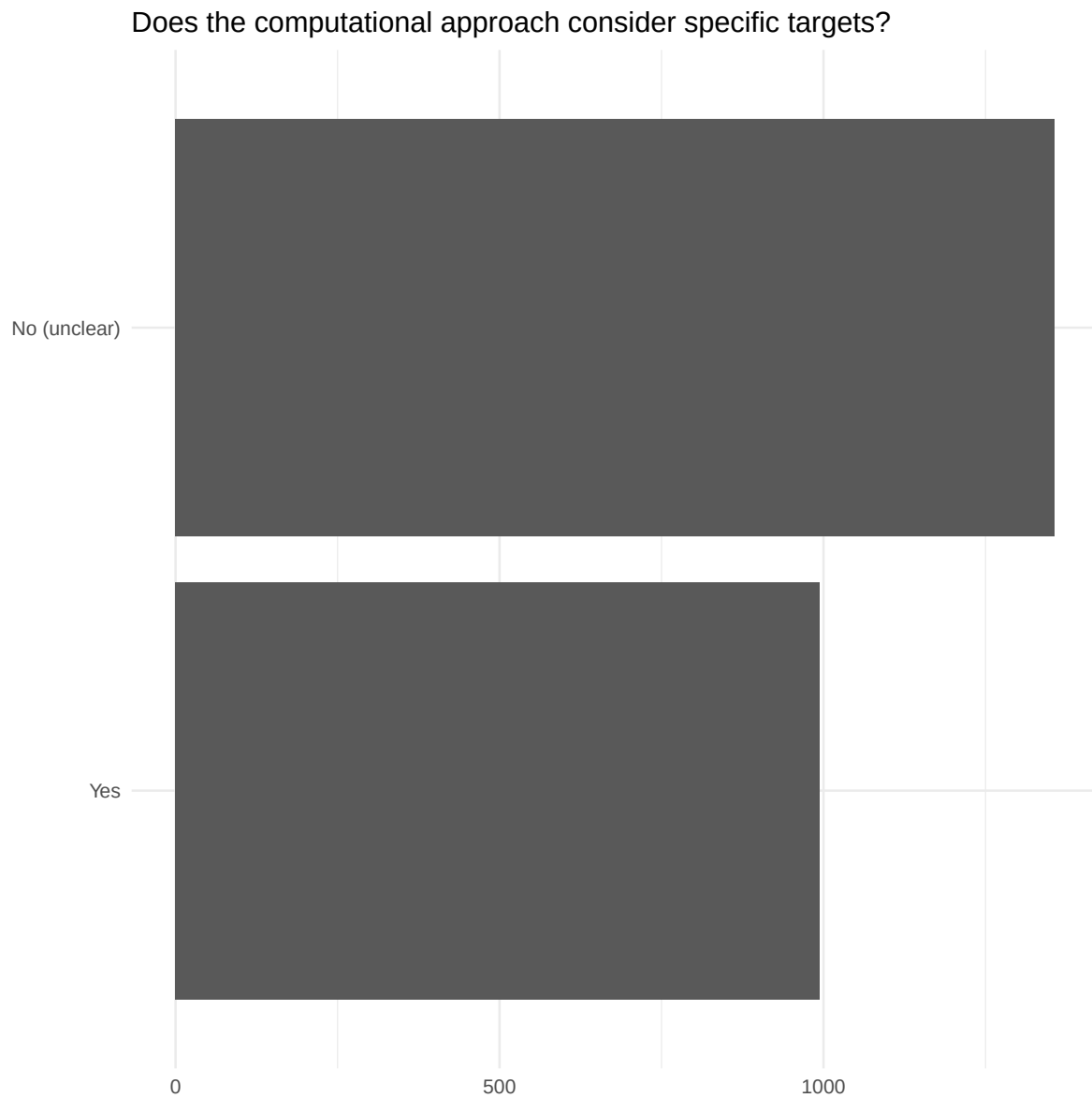


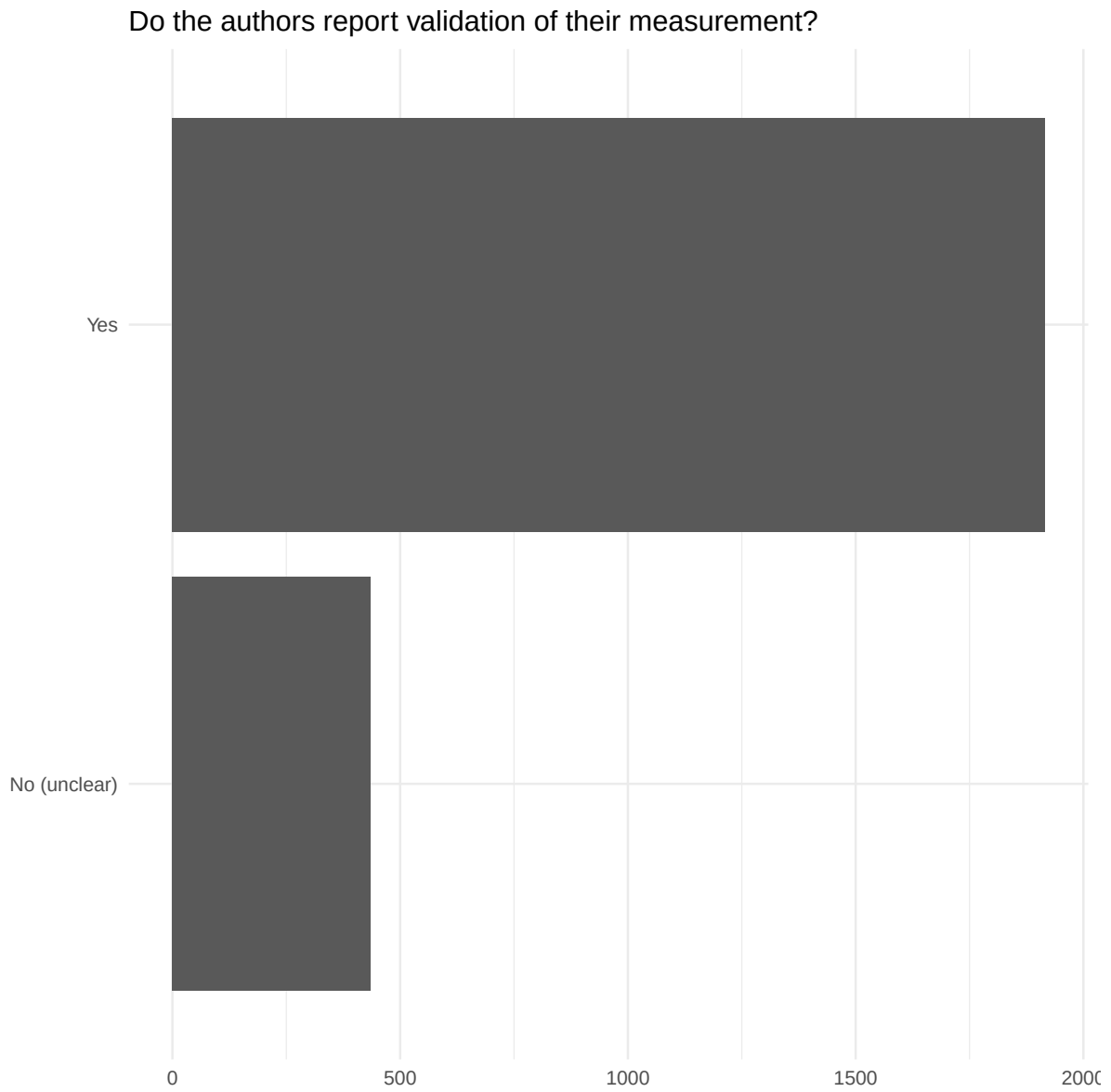
Figure 2: Approaches used for Opinion Measurement

## RQ1d: Which data is used in CSS opinion mining?

### Target



## Validation



## RQ2: disciplinary differences

concepts

methods

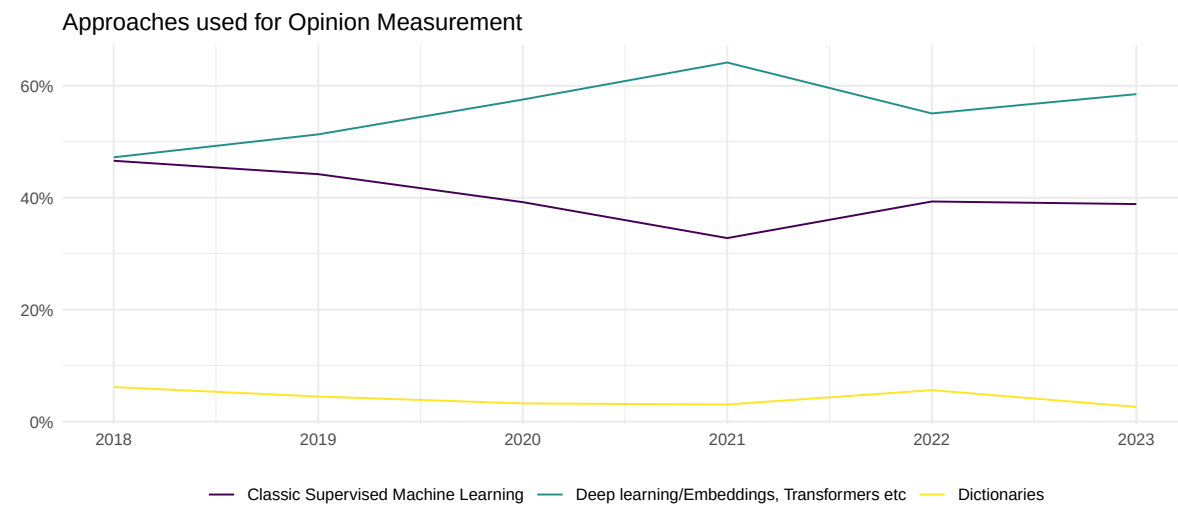
tools

data

## RQ3: change over time

concepts

methods



tools

data

## wrap up

Save data if it takes long to preprocess it here. Afterwards we get some information which is important to reproduce the report.

R version 4.5.2 (2025-10-31)  
Platform: x86\_64-pc-linux-gnu  
Running under: EndeavourOS

Matrix products: default  
BLAS: /usr/lib/libblas.so.3.12.0  
LAPACK: /usr/lib/liblapack.so.3.12.0 LAPACK version 3.12.0

locale:

[1] LC\_CTYPE=en\_GB.UTF-8 LC\_NUMERIC=C  
[3] LC\_TIME=en\_GB.UTF-8 LC\_COLLATE=en\_GB.UTF-8  
[5] LC\_MONETARY=en\_GB.UTF-8 LC\_MESSAGES=en\_GB.UTF-8  
[7] LC\_PAPER=en\_GB.UTF-8 LC\_NAME=C  
[9] LC\_ADDRESS=C LC\_TELEPHONE=C  
[11] LC\_MEASUREMENT=en\_GB.UTF-8 LC\_IDENTIFICATION=C

time zone: Europe/Berlin  
tzcode source: system (glibc)

attached base packages:

[1] stats graphics grDevices utils datasets methods base

other attached packages:

[1] lubridate\_1.9.4 forcats\_1.0.0 stringr\_1.6.0 dplyr\_1.2.0  
[5] purrr\_1.2.1 readr\_2.1.5 tidyr\_1.3.1 tibble\_3.3.1  
[9] ggplot2\_3.5.2 tidyverse\_2.0.0

loaded via a namespace (and not attached):

[1] gtable\_0.3.6 jsonlite\_2.0.0 compiler\_4.5.2 tidyselect\_1.2.1  
[5] scales\_1.4.0 yaml\_2.3.12 fastmap\_1.2.0 here\_1.0.2  
[9] R6\_2.6.1 labeling\_0.4.3 generics\_0.1.4 curl\_7.0.0  
[13] knitr\_1.51 rprojroot\_2.1.1 pillar\_1.11.1 RColorBrewer\_1.1-  
3  
[17] tzdb\_0.5.0 R.utils\_2.13.0 rlang\_1.1.7 stringi\_1.8.7  
[21] xfun\_0.55 viridisLite\_0.4.2 timechange\_0.3.0 cli\_3.6.5  
[25] withr\_3.0.2 magrittr\_2.0.4 rio\_1.2.3 digest\_0.6.39  
[29] grid\_4.5.2 hms\_1.1.3 lifecycle\_1.0.5 R.oo\_1.27.0  
[33] R.methodsS3\_1.8.2 vctrs\_0.7.1 data.table\_1.17.0 evaluate\_1.0.5  
[37] glue\_1.8.0 farver\_2.1.2 rmarkdown\_2.30 tools\_4.5.2  
[41] pkgconfig\_2.0.3 htmltools\_0.5.9

[1] "2026-02-09 17:34:06 CET"

Time difference of 36.56732 secs